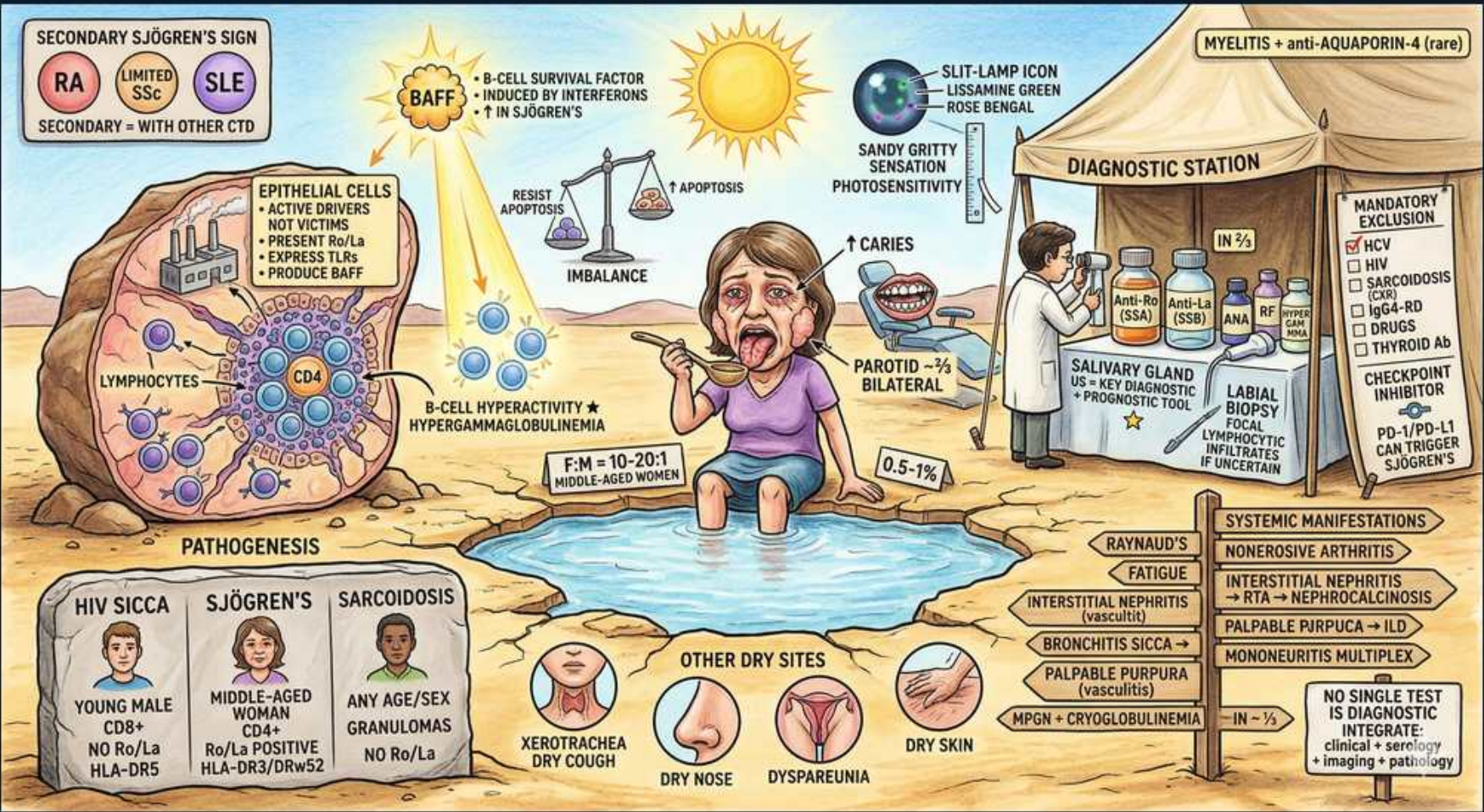


Sjögren — Pathogenesis & Diagnosis



1. Sicca (dry eyes/mouth)

WHAT YOU SEE

A weary middle-aged woman sitting in a cracked, bone-dry lakebed — the dried-up lake IS the dried-up exocrine glands, no tears and no saliva left.

WHAT IT ENCODES

Sjögren is a chronic autoimmune exocrinopathy: CD4 lymphocytes infiltrate and destroy the lacrimal and salivary glands, producing the sicca complex — keratoconjunctivitis sicca (dry, gritty eyes) and xerostomia (dry mouth). Patients describe needing water to swallow dry food, burning eyes, and a 'sand in the eyes' sensation. The dried lakebed is the mnemonic: the glands have run dry. Primary Sjögren occurs alone; secondary Sjögren accompanies another connective tissue disease (most often RA, then SLE/scleroderma).

BOARD TRAP

WRONG: attributing isolated dry eyes/mouth in an older patient to Sjögren without objective testing. Anticholinergic medications (TCAs, antihistamines, diuretics), age-related glandular atrophy, prior head/neck radiation, and uncontrolled diabetes all cause sicca symptoms. Discriminator: true Sjögren has objective findings (abnormal Schirmer/ocular staining, focal lymphocytic sialadenitis on biopsy, anti-Ro/La) — symptomatic dryness alone is not enough.

2. BAFF / B-cell survival

WHAT YOU SEE

A blazing sun labeled BAFF / B-CELL SURVIVAL FACTOR — the sun's rays keep autoreactive B cells alive (rescued from apoptosis), driving the disease.

WHAT IT ENCODES

BAFF (B-cell activating factor, also called BLyS) is induced by type I interferons in Sjögren and acts as a survival signal that rescues autoreactive B cells from apoptosis. Elevated BAFF drives the B-cell hyperactivity, hypergammaglobulinemia, and autoantibody production central to the disease. This is the same cytokine targeted by belimumab in SLE, and the interferon/BAFF axis explains the autoantibody-rich phenotype.

BOARD TRAP

WRONG: thinking Sjögren is primarily a T-cell cytotoxic disease like HIV-associated DILS. Discriminator: although CD4 T cells initiate glandular infiltration, the BAFF-driven B-cell arm produces the autoantibodies and underlies the markedly increased lymphoma risk — it is fundamentally a B-cell–hyperactive disease.

3. B-cell hyperactivity

WHAT YOU SEE

A banner reading B-CELL HYPERACTIVITY / HYPERGAMMAGLOBULINEMIA — the overworked B cells flooding out antibodies, the same overdrive that fuels MALT lymphoma risk.

WHAT IT ENCODES

Sustained B-cell hyperactivity produces polyclonal hypergammaglobulinemia, rheumatoid factor positivity, cryoglobulins, and the anti-Ro/La antibodies. Critically, this chronic B-cell drive carries a 5–15× increased risk of non-Hodgkin lymphoma (typically MALT lymphoma of the parotid). Red flags for transformation: persistent unilateral parotid enlargement, new lymphadenopathy, low C4, cryoglobulinemia, and a falling/disappearing previously-high IgG with a new monoclonal band.

BOARD TRAP

WRONG: reassuring a Sjögren patient with persistent firm unilateral parotid swelling that it is just a flare. Discriminator: bilateral soft glandular swelling is typical Sjögren, but a persistent firm unilateral mass, new low C4, or cryoglobulinemia must prompt biopsy to exclude MALT lymphoma — the most important long-term complication.

4. Anti-Ro/SSA

WHAT YOU SEE

A bottle labeled ANTI-Ro/SSA on the diagnostic station shelf — the key Sjögren antibody, also the one that crosses the placenta to cause neonatal heart block.

WHAT IT ENCODES

Anti-Ro/SSA is the key Sjögren autoantibody (positive in ~70%) and contributes 3 points toward the 2016 ACR/EULAR classification. It is also responsible for neonatal lupus and congenital complete heart block (anti-Ro crosses the placenta and damages the fetal AV node ~18–24 weeks), so pregnant Ro-positive women need fetal echo monitoring. Anti-Ro can occur in SLE and SCLE as well, so it is not specific to Sjögren.

BOARD TRAP

WRONG: assuming a young woman with anti-Ro and dry eyes has a benign condition with no pregnancy implications. Discriminator: anti-Ro/SSA positivity mandates counseling about congenital heart block and neonatal lupus risk and serial fetal monitoring in pregnancy — a high-yield board association.

5. Anti-La/SSB

WHAT YOU SEE

A bottle labeled ANTI-La/SSB beside the Ro bottle — La almost always travels WITH Ro, the classic 'Ro and La = Sjögren' pairing.

WHAT IT ENCODES

Anti-La/SSB almost always accompanies anti-Ro and supports the diagnosis; isolated anti-La (without anti-Ro) is rare and of uncertain significance. The classic high-yield pairing is 'Ro and La = Sjögren,' though both also appear in SLE. La positivity correlates with a more complete sicca phenotype.

BOARD TRAP

WRONG: relying on isolated anti-La to diagnose Sjögren. Discriminator: anti-La rarely occurs alone and adds little diagnostic weight without anti-Ro; the 2016 criteria give points to anti-Ro/SSA, not isolated anti-La.

6. Schirmer / rose bengal

WHAT YOU SEE

A slit-lamp icon with ROSE BENGAL and LISSAMINE GREEN dye plus a SANDY GRITTY SENSATION note — the dyes stain the devitalized dry-eye surface, proving objective sicca.

WHAT IT ENCODES

Objective ocular testing confirms aqueous tear deficiency: the Schirmer test (filter paper in the lower lid) is abnormal at ≤5 mm of wetting in 5 minutes. Ocular surface staining with rose bengal/lissamine green (ocular staining score ≥5) or fluorescein reveals devitalized epithelium from chronic dryness. These objective findings are required because symptomatic dryness is nonspecific.

BOARD TRAP

WRONG: diagnosing Sjögren on symptoms alone without objective ocular or salivary testing. Discriminator: the classification criteria require objective evidence — Schirmer ≤5 mm/5 min, ocular staining score ≥5, or unstimulated salivary flow ≤0.1 mL/min — distinguishing true Sjögren from drug-induced or age-related dryness.

7. Labial gland biopsy

WHAT YOU SEE

The LABIAL/SALIVARY GLAND biopsy station noting FOCAL LYMPHOCYTIC SIALADENITIS — the gold-standard tissue proof (focus score ≥1), worth the same points as anti-Ro.

WHAT IT ENCODES

Minor (labial) salivary gland biopsy is the gold-standard histologic confirmation. The diagnostic finding is focal lymphocytic sialadenitis with a focus score ≥1 (≥1 focus of ≥50 lymphocytes per 4 mm²). This carries 3 points in the 2016 ACR/EULAR criteria — equal weight to anti-Ro. Biopsy is most useful in seronegative (anti-Ro–negative) patients with sicca to secure the diagnosis.

BOARD TRAP

WRONG: ordering a labial biopsy first-line in a clearly anti-Ro–positive patient with objective sicca. Discriminator: biopsy is reserved for diagnostic uncertainty, especially anti-Ro–negative cases; a focus score ≥1 confirms it, whereas nonspecific chronic sialadenitis without focal aggregates does not count.

8. Bilateral parotid

WHAT YOU SEE

A PAROTID label marked BILATERAL over both cheeks — soft symmetric swelling is benign Sjögren; the board pivot is that a firm UNILATERAL mass means lymphoma.

WHAT IT ENCODES

Recurrent bilateral parotid (and submandibular) gland enlargement is characteristic of Sjögren, reflecting lymphocytic infiltration. It is typically symmetric, soft, and fluctuating. The key board pivot: persistent firm UNILATERAL enlargement signals possible MALT lymphoma and warrants imaging and biopsy.

BOARD TRAP

WRONG: equating any parotid swelling with active Sjögren. Discriminator: bilateral, soft, episodic swelling = benign Sjögren; persistent unilateral firm swelling = lymphoma until proven otherwise. Also distinguish from mumps, sarcoidosis, and HIV (DILS), which cause bilateral parotid enlargement without anti-Ro/La.

9. Caries

WHAT YOU SEE

CARIES marked at the woman's mouth — rampant gum-line cavities because protective saliva is gone, a classic dry-mouth clue.

WHAT IT ENCODES

Loss of protective, antibacterial saliva causes rampant dental caries (often at the gum line and tooth necks), oral candidiasis (angular cheilitis, atrophic glossitis), and difficulty wearing dentures. Aggressive accelerated caries in an adult with dry mouth is a classic clue. Management includes meticulous dental care, fluoride, sialogogues (pilocarpine/cevimeline), and saliva substitutes.

BOARD TRAP

WRONG: overlooking accelerated multi-surface caries as a Sjögren clue or treating only the teeth. Discriminator: rampant caries plus xerostomia should prompt evaluation for Sjögren (anti-Ro/La, Schirmer, salivary flow) — and sialogogues like pilocarpine/cevimeline target the underlying hyposalivation.

10. Female 9:1 middle-aged

WHAT YOU SEE

A label reading F:M 9-10:1, MIDDLE-AGED WOMEN — the strong female, midlife demographic that separates true Sjögren from the young-male HIV mimic.

WHAT IT ENCODES

Sjögren has a striking female predominance (~9–10:1) and typically presents in middle-aged women (40s–50s), the same demographic as most systemic autoimmune disease. This epidemiology is high-yield and helps separate true Sjögren from the HIV/DILS mimic (young male).

BOARD TRAP

WRONG: anchoring on demographics to exclude HIV-associated sicca. Discriminator: a YOUNG MALE with sicca and bilateral parotid swelling but NEGATIVE anti-Ro/La is HIV/DILS (CD8-driven), not classic middle-aged-female Sjögren — check an HIV test.

11. HIV sicca mimic

WHAT YOU SEE

A boxed mimic card: HIV SICCA — YOUNG MALE, CD8, NO Ro/La, HLA-DR5 — the CD8-driven DILS look-alike, mechanistically opposite to Sjögren.

WHAT IT ENCODES

Diffuse infiltrative lymphocytosis syndrome (DILS) in HIV mimics Sjögren with sicca and parotid enlargement, but is mechanistically opposite: CD8 (not CD4) lymphocytic infiltration, typically a young male, anti-Ro/La NEGATIVE, and associated with HLA-DR5. The board lesson is to test for HIV in atypical sicca.

BOARD TRAP

WRONG: labeling a young man with sicca, parotid swelling, and negative autoantibodies as primary Sjögren. Discriminator: CD8-predominant infiltrate, male sex, young age, absent anti-Ro/La, and HLA-DR5 point to HIV/DILS — order an HIV test before diagnosing Sjögren.

12. Sjögren CD4 / HLA-DR3-DR52

WHAT YOU SEE

A boxed reference card: SJÖGREN — MIDDLE-AGED WOMAN, CD4, Ro/La POSITIVE, HLA-DR3/DRw52 — the true-disease phenotype the mimics are contrasted against.

WHAT IT ENCODES

True primary Sjögren is CD4 T-cell driven, occurs in middle-aged women, is anti-Ro/La positive, and is HLA-associated with DR3/DRw52. This is the reference phenotype against which the HIV (CD8, DR5) and sarcoid (granulomas, Ro/La negative) mimics are contrasted on the board.

BOARD TRAP

WRONG: confusing the immunogenetics of the mimics. Discriminator: Sjögren = CD4 + Ro/La positive + HLA-DR3/DRw52; HIV-DILS = CD8 + Ro/La negative + HLA-DR5; sarcoid = noncaseating granulomas + Ro/La negative. The CD4-vs-CD8 and Ro/La status are the cleanest discriminators.

13. Sarcoid sicca mimic

WHAT YOU SEE

A boxed mimic card: SARCOIDOSIS — ANY AGE/SEX, GRANULOMAS, NO Ro/La — the granulomatous gland infiltrate look-alike (biopsy shows granulomas, not lymphocytes).

WHAT IT ENCODES

Sarcoidosis causes sicca and parotid enlargement through noncaseating granulomatous infiltration of glands (and can produce Heerfordt syndrome: parotid enlargement, uveitis, facial palsy, fever). It occurs at any age/sex and is anti-Ro/La NEGATIVE; biopsy shows granulomas, not focal lymphocytic sialadenitis. Hypercalcemia and elevated ACE may be present.

BOARD TRAP

WRONG: diagnosing Sjögren when a labial/parotid biopsy shows noncaseating granulomas. Discriminator: granulomas (not lymphocytic foci) plus negative anti-Ro/La, possible hypercalcemia/elevated ACE, and hilar adenopathy point to sarcoidosis, not Sjögren.

14. Renal tubular acidosis / nephritis

WHAT YOU SEE

The systemic-manifestations list including INTERSTITIAL NEPHRITIS and TYPE 1 RTA — Sjögren attacks the tubulointerstitium, giving distal RTA, not glomerulonephritis.

WHAT IT ENCODES

Sjögren's classic renal lesion is tubulointerstitial nephritis (lymphocytic infiltration of the interstitium), causing distal (type 1) renal tubular acidosis. Type 1 RTA presents with a non-anion-gap metabolic acidosis, inappropriately alkaline urine (pH >5.5), hypokalemia (can cause periodic paralysis), and nephrocalcinosis/calcium phosphate stones. Glomerulonephritis is uncommon and, when present, is often associated with cryoglobulinemia.

BOARD TRAP

WRONG: expecting glomerulonephritis with nephrotic-range proteinuria as the typical Sjögren kidney disease. Discriminator: Sjögren renal involvement is predominantly INTERSTITIAL with distal (type 1) RTA — non-anion-gap acidosis, urine pH >5.5, hypokalemia — not the proliferative glomerulonephritis seen in lupus nephritis.

15. Other dry sites

WHAT YOU SEE

Icons for XEROTRACHEA/DRY COUGH, DRY NOSE, DYSPAREUNIA, and DRY SKIN — every mucosal surface dries out, showing Sjögren is a systemic exocrinopathy.

WHAT IT ENCODES

Glandular dryness extends beyond eyes and mouth: xerotrachea causes a chronic dry cough; nasal dryness causes crusting/epistaxis; vaginal dryness causes dyspareunia; and skin becomes xerotic. Recognizing multi-site mucosal dryness reinforces that Sjögren is a systemic exocrinopathy, and these symptoms guide supportive care (humidification, vaginal moisturizers, emollients).

BOARD TRAP

WRONG: dismissing chronic dry cough in a sicca patient as unrelated or as asthma/GERD. Discriminator: in established Sjögren, xerotrachea is a direct manifestation of the exocrinopathy — but new dyspnea/infiltrates should still prompt evaluation for interstitial lung disease, a recognized systemic complication.
